Develop Sequence and Collaboration Diagram

Behavioral model

Creating a behavioral model

The behavioral model indicates how software **will respond to external events**.

To create the model, **perform the following steps**:

1. Evaluate all use cases to fully understand the sequence of interaction within the system.
2. Identify events that drive the interaction sequence and understand how these events relate to specific objects.
3. Create a sequence for each use case.
4. Build a state diagram for the system.
5. Review the behavioral model to verify accuracy and consistency.

Sequence diagrams

- Sequence diagram is **behavioral representation** which indicates **how events cause transitions** from **object to object.**
- It is the most commonly used **interaction** diagram.
- Once events have been identified by examining a use case, the modeler creates a sequence diagram—a representation of **how events cause flow from one object to another as a function of time.**
- The sequence diagram **represents key classes/objects and the events that cause behavior to flow from class to class.**

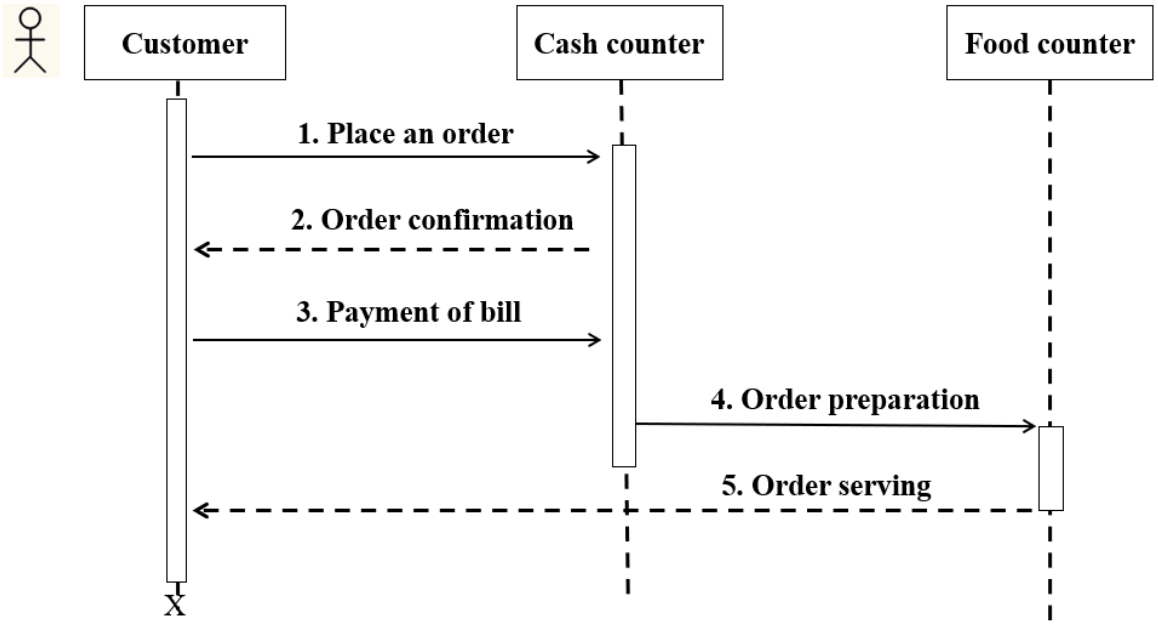
Example – fast food order system

Nouns which may help to identify classes/objects can be

- Customer
- Payment &
- Order

Events taking place and passing/flowing from one object to another at a different time are

1. Place an order
2. Order confirmation
3. Payment of bill
4. Order preparation &
5. Order serving



Sequence Diagram Notations

Actors –

Actor initiates action, who is always outside the scope of the system.

Actor can be represented as,



Object –

An object is **a named element which depicts an individual participant** in a sequence diagram.

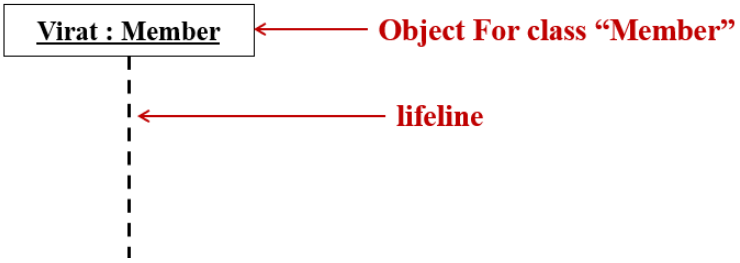
Basically, **each instance in a sequence diagram** is represented by a **object**.

An **object is always internal to the system**.

Objects are displayed with a **rectangle** called **head** with its **name** and **class type**.

The object head is located on top of a **vertical dashed line** (referred to as a **lifeline**).

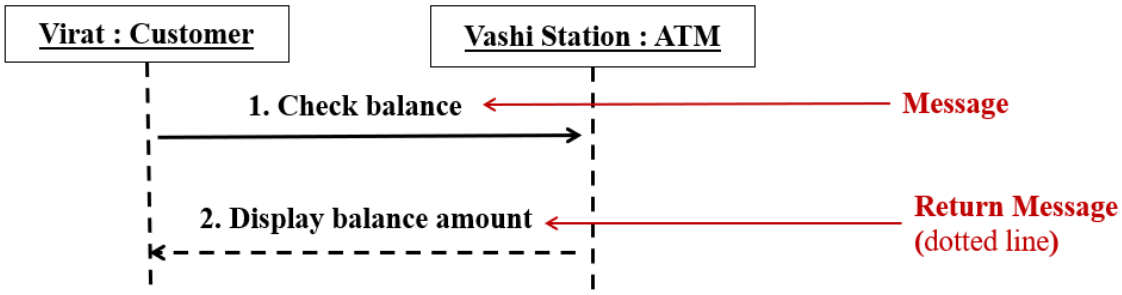
Example - For class "Member" of library management system, object Virat can be represented as



Messages -

Communication between objects is depicted using messages.

The messages **appear along with arrows in a sequential order on the lifeline**.

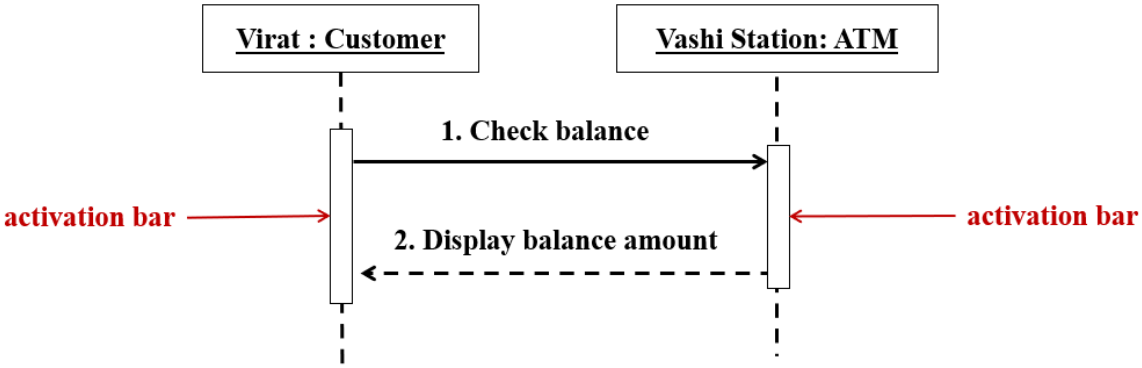


Each message has caller (the object that sends the message) and the message receiver (the object that receives the message)

Numbering of messages is **optional** as the messages appear along with arrows in a sequential order on the lifeline, but it's **preferred**.

Activation bar-

The use of the **activation bar** on the lifelines of the message caller and the message receiver indicates that **both are active during the exchange of the message.**



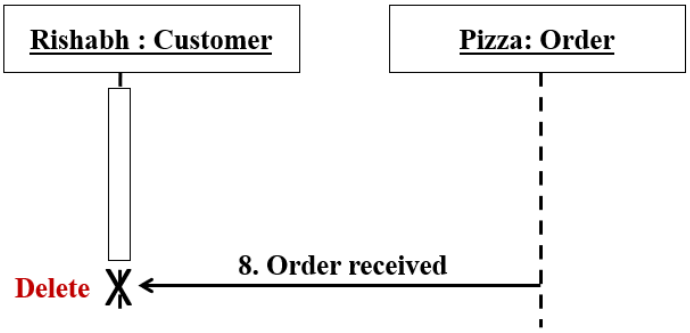
Delete Message -

Delete Message is used to **delete an object**.

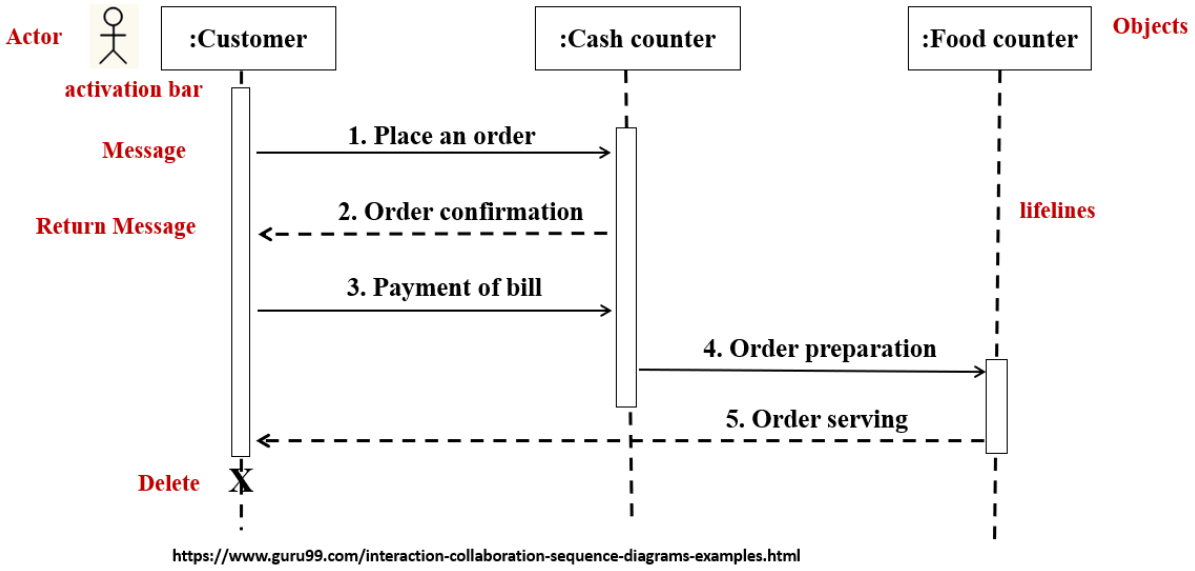
It **destroys the occurrence of the object in the system**.

It is represented by an **arrow terminating with a X**.

For example - when the order is received by the user, the object of user class can be destroyed.



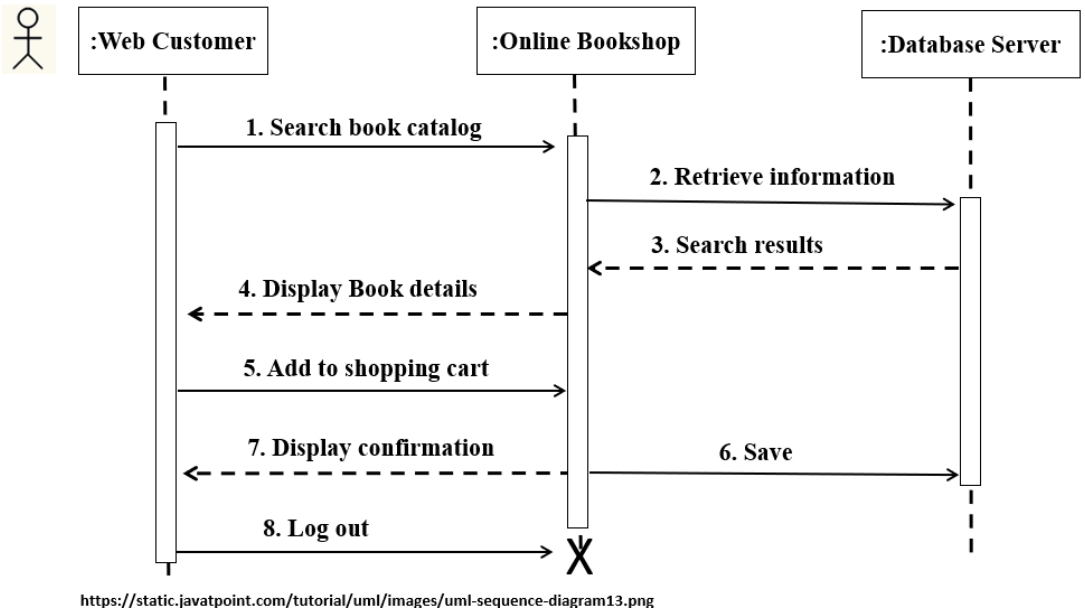
Example – Sequence diagram for fast food order system



https://www.guru99.com/interaction-collaboration-sequence-diagrams-examples.html

Example – Sequence diagram for online bookshop

Any online customer can search for a book catalog, view a description of a particular book, add a book to its shopping cart, and do checkout.

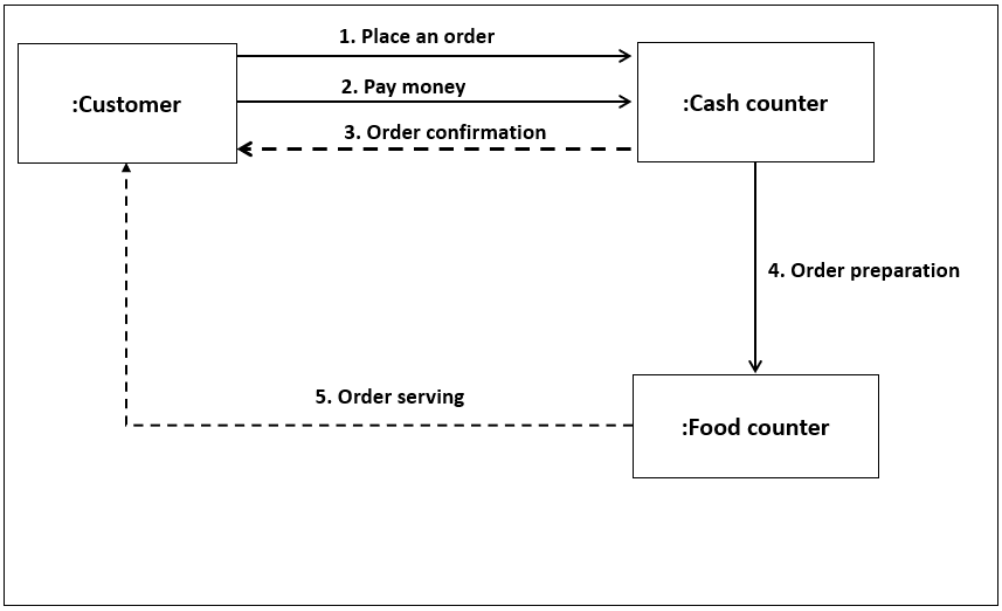


https://static.javatpoint.com/tutorial/uml/images/uml-sequence-diagram13.png

UML Collaboration Diagram

- The collaboration diagram is used **to show the relationship between the objects in a system**.
- Both the sequence and the collaboration diagrams **represent the same information but differently**.
- Instead of showing the flow of messages, it depicts the architecture of the object residing in the system as it is based on object-oriented programming.
- The collaboration diagram is also known as a **communication diagram.**

Example- Collaboration Diagramfor fast food order system



Numbering of messages is **mandatory** as messages may not be in a **sequential order** .

Example – Sequence diagram for online bookshop

